

WIRELESS DISPLAY Using Arduino

RUSHIRAJ S. JAWALE

Arduino Uno is a microcontroller board based on ATmega328. This DIY demonstrates how you can use two Arduino boards to make a wireless display using 434MHz ASK RF modules.

Data is transmitted wirelessly using the RF module from the trans-

mitter end through the serial monitor of Arduino IDE. It is received at the receiver end and displayed on a common 16x2 LCD.

This system can be used as wireless display for temperature, humidity and gas sensors. It has a good range and, using a repeater, the range can be extended.

Circuit and working

Arduino Uno board is used in both transmitter and receiver circuits. Let us name the transmitting Arduino as Arduino1 and the receiving Arduino as Arduino2. You can use any Arduino board that has ATmega328.

Circuit diagram of the transmitter is shown in Fig. 1. You need VirtualWire library in Arduino to make the circuit work.

In Fig. 1, digital pin 12 of Arduino1 is used to

transmit the data serially. This serial data is fed to the ASK RF transmitter module (TX1). ASK stands for amplitude shift keying, which means that amplitude of the pulse changes according to the serial data. It is a form of digital modulation.

Since data is sent using the serial monitor in Arduino IDE, Arduino1 needs to be always connected to a PC/laptop. Enter the data to be sent in the serial monitor followed by a full stop (.) and press Enter. (Please note that the full stop is important. Data will not be sent if it is not terminated with a full stop.)

PARTS LIST

Semiconductors:

- BOARD1, BOARD2 - Arduino Uno board

- Resistors (all 1/4-watt, 5% carbon):
- VR1 - 10-kilo-ohm preset

Miscellaneous:

- LCD1 - 16x2 alphanumeric display
- ASK RF Module - ASK RF module pair (TX1, RX1)
- ANT.1, ANT.2 - 17cm long single-strand wire antenna

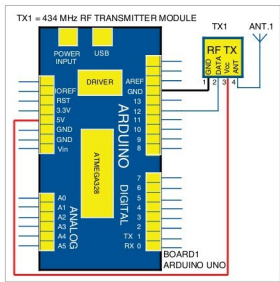


Fig. 1: Transmitter circuit (Arduino1)

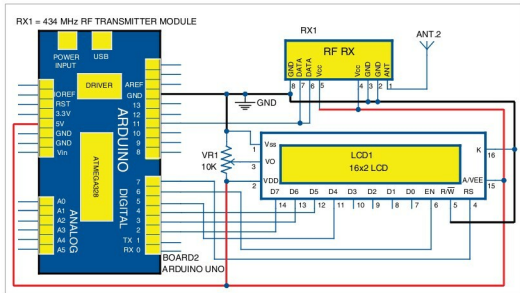


Fig. 2: Receiver circuit (Arduino2)

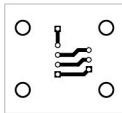


Fig. 3: An actual-size PCB pattern of the transmitter circuit

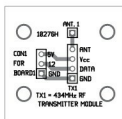


Fig. 4: Component layout of the transmitter PCB